Śrīmad Bhāgavatam Chitra-Katha

Kṛṣṇa Meets Kubjā

Student Name: Prajjwal Patel
Roll Number: 24155637
Course: Speech and Video Processing
Source: Śrīmad Bhāgavatam, Canto 10, Chapter 42, verses 1–12

Abstract

This project specifies a reproducible multimodal storytelling pipeline. Media results depend on user-supplied narration and illustrations; absent media are reported as incomplete.

Introduction

The episode is retold in twelve child-friendly panels while distinguishing the chapter's narrative from creative staging and dialogue.

Objectives

Create comic layouts, demonstrate speech filtering and analysis, generate synchronized video, measure real outputs, and document limitations.

Source narrative and references

Primary source: https://vedabase.io/en/library/sb/10/42/ (Canto 10, Chapter 42, verses 1–12). Status: verified_by_online_chapter_check.

System architecture

JSON story and source manifest feed Pillow comic layout, SciPy audio processing, OpenCV motion, FFmpeg encoding, evaluation, and ReportLab.

Speech methodology

Input WAV is converted to mono 44.1 kHz by polyphase resampling. A second-order high-pass filter is demonstrated. No noise-reduction/SNR benefit is presumed.

Video and synchronization

Panels receive deterministic zoom/pan motion. SRT durations are estimated from word count over measured audio duration. This is an estimate, not forced alignment; manual verification is required.

Quantitative evaluation

Only metrics derived from supplied audio/video are written. Missing assets result in unavailable metrics, never fabricated values.

Source vs Adaptation

Source: Trivakrā offers ointment; Kṛṣṇa straightens her; she invites Him; He promises to visit after His purpose. Adaptation: simplified dialogue, scenic backgrounds, camera motion, transitions, reflection, and restrained child-friendly framing.

Children's reflection

We can offer what we have with generosity, speak kindly, and treat every person with dignity. A person's worth is never measured by appearance.

Limitations

No recording or finished illustration assets are bundled. FFmpeg is an external prerequisite. This environment may lack Python/media dependencies. Timing estimates need human review.

Conclusion

The code separates source claims from creative presentation and supports measured processing when actual media is supplied.

Problem statement

An accurate illustrated retelling requires coordinated speech analysis, video assembly, timing, and source attribution.

Comic design and panel mapping

Twelve child-friendly panels map to chapter verses 1-12 in the source manifest. User-supplied artwork is required for finished illustrations.

Waveform and spectrogram analysis

Waveforms show amplitude over time. Spectrograms use STFT, NFFT 1024, hop 256, Hann window, dB scale, and frequency display to 10 kHz.

Noise reduction and filtering

A second-order high-pass filter is demonstrated. It can alter voice fundamentals; no SNR benefit is claimed without a noise-only sample.

Video processing and frame extraction

Deterministic zoom/pan is rendered, then frames are sampled from the encoded MP4 at configured intervals.

References

Bhaktivedanta Book Trust International, Srimad Bhagavatam 10.42: https://vedabase.io/en/library/sb/10/42/ .

Appendix: project structure and execution

See README.md for the project structure, setup, commands, required WAV and panel assets, and troubleshooting.

Measured results

audio_metrics.json

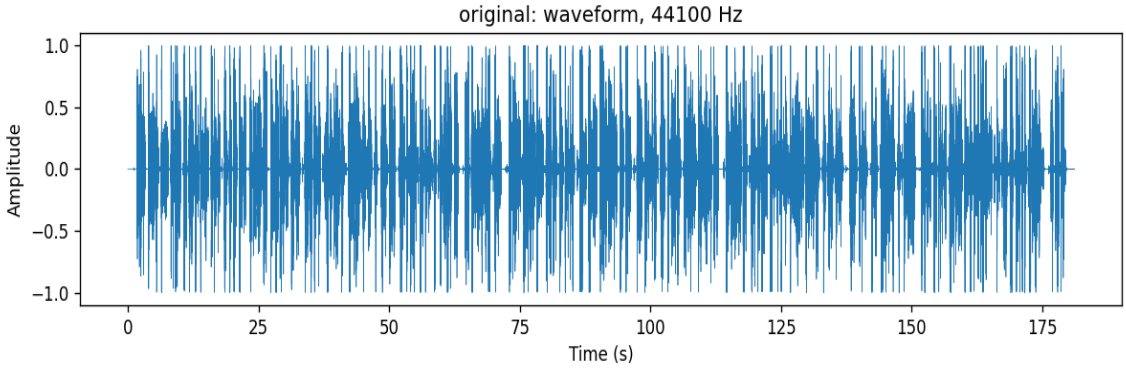
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video_metrics.json

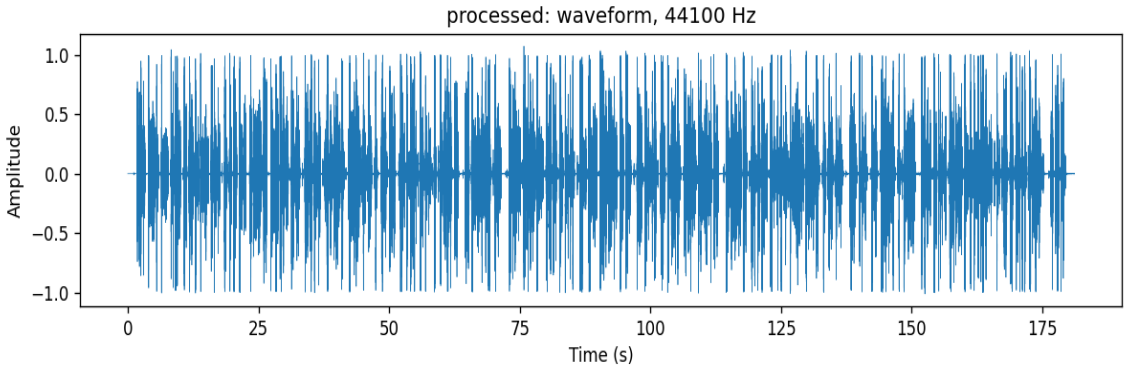
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evaluation.json

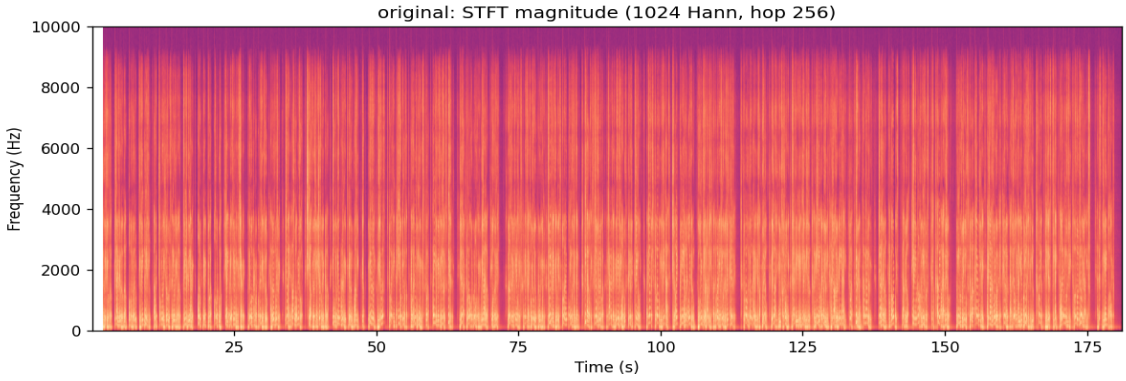
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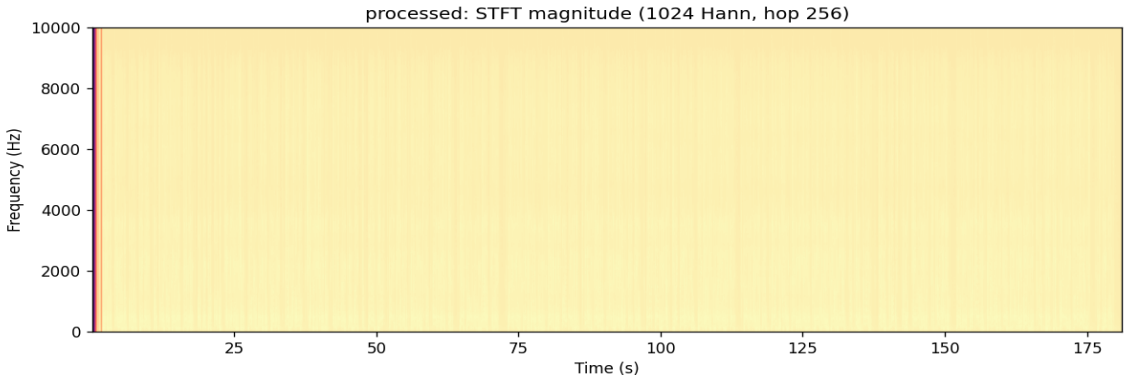
waveform_original.png



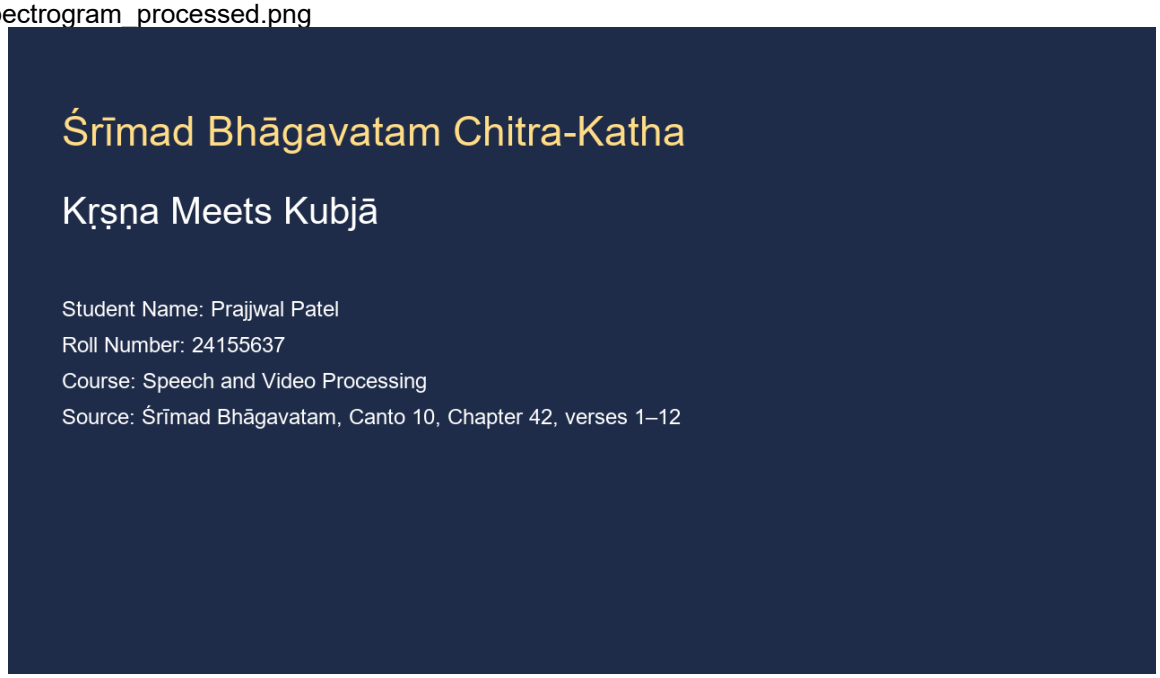
waveform_processed.png



spectrogram_original.png



spectrogram_processed.png



Bhagavatam says / Our visual adaptation adds

Bhagavatam says	Our visual adaptation adds
Verses 1-5: Trivakra meets the brothers and offers unguents.	Mathura street scenery, simplified dialogue and decorative framing.
Verses 6-8: Krishna straightens Trivakra.	Gentle camera movement and respectful, non-sensational framing.
Verses 9-12: She invites Krishna; He says He will visit after fulfilling His purpose.	Simplified dialogue, transitions and a children's reflection.

Panel-to-source map

Verse	Narrative fact	Panel	Adaptation
1	Kṛṣṇa meets Trivakrā with fragrant ointments in the royal road.	P01–P02	Mathura street scenery and framing.
2	Kṛṣṇa asks about the ointments and requests some for Himself and Balarāma.	P03	Simplified dialogue, not quotation.
3	She serves Kaṁsa making ointments and offers them to the brothers.	P04	Simplified dialogue.
4	Trivakrā generously gives ointments to Kṛṣṇa and Balarāma.	P05	Visual framing.
5	The brothers are adorned with ointment and appear beautiful.	P06	Background colors are decorative additions.
6	Kṛṣṇa is pleased and decides to straighten Trivakrā.	P07	Compassionate emphasis is interpretive narration.
7	Kṛṣṇa presses her toes with His feet and lifts her chin, straightening her body.	P07	Gentle, straightening is a bodily choice.
8	Trivakrā is transformed with straight, proportionate limbs.	P08	Non-sensational depiction; worth is not tied to appearance.
9	Trivakrā takes hold of Kṛṣṇa's upper cloth and addresses Him, longing.	P09	Gestures sensitive, longing summarized for children.
10	Trivakrā invites Kṛṣṇa to her home.	P09	Simplified dialogue, not quotation.
11	Kṛṣṇa glances at Balarāma and the cowherd boys, smiling.	P10	Eyes looking, replying.
12	Kṛṣṇa says He will visit after fulfilling His purpose.	P10–P11	Simplified dialogue, not quotation.

Execution

Run python -m src.pipeline --config config.yaml --stage all. Provide audio/original/narration.wav and assets/panels/panel_01.png through panel_12.png for media generation.